First-Order and Higher-Order Differential Equations

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§1.1 First-Order ODEs and Substitutions

A differential equation relates an unknown function to its derivatives. The order is the highest derivative appearing. A general first-order equation has the form

$$F(x, y, y') = 0.$$

A solution is a function $y(x)$ satisfying the equation on an interval. When the coefficients and regularity conditions support existence and uniqueness, initial data select a particular solution from a family with integration constants. The development follows the MIT 18.03 *Differential Equations* lectures and Arthur Mattuck's supplementary notes on operators and impulse responses.[1] Initial conditions alone do not guarantee uniqueness: the equation $y' = 2\sqrt{|y|}$, $y(0) = 0$ has both $y \equiv 0$ and a solution that stays zero for $t \le c$ and then equals $(t - c)^2$ for $t > c$, for any $c \ge 0$.

A separable equation has

$$\frac{dy}{dx} = g(x)h(y).$$

If $h(y) \neq 0$, then

$$\frac{dy}{h(y)} = g(x)\,dx.$$

Integrating gives an implicit solution. One must also check equilibrium solutions coming from $h(y) = 0$, because division by $h(y)$ can lose them.

A homogeneous first-order equation has the form

$$y' = F\left(\frac{y}{x}\right).$$

Set $v = y/x$, so $y = vx$ and

$$y' = v + xv'.$$

Then

$$v + x\frac{dv}{dx} = F(v),$$

which is separable:

$$\frac{dv}{F(v) - v} = \frac{dx}{x}.$$

Some equations become homogeneous after translating the origin. If the linear expressions in x, y meet at a point, shift that point to the origin.

[1] https://ocw.mit.edu/courses/18-03-differential-equations-spring-2010/pages/readings/

§1.2 Integrating Factors and Exact Forms

A first-order linear equation is

$$y' + p(x)y = q(x).$$

The integrating factor is

$$\mu(x) = e^{\int p(x)\,dx}.$$

Multiplying the equation by μ gives

$$(\mu y)' = \mu q.$$

Thus

$$y = e^{-\int p}\left(\int q e^{\int p}\,dx + C\right).$$

The integrating factor is not a trick; it conjugates the operator $D+p(x)$ into the ordinary derivative D.

A Bernoulli equation has

$$y' + p(x)y = q(x)y^n.$$

For $n \neq 0, 1$, set

$$u = y^{1-n}.$$

Then

$$u' = (1-n)y^{-n}y',$$

and the equation becomes linear in u. Again, check special solutions lost by division. For example $y' + y = y^2$ becomes, on intervals where $y \neq 0$, $u' - u = -1$ under $u = 1/y$. Thus $u = 1 + Ce^x$ and $y = (1 + Ce^x)^{-1}$, but the equilibrium solution $y \equiv 0$ was lost during division and must be added separately.

An equation

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

is exact if there is a potential Φ with

$$d\Phi = M\,dx + N\,dy.$$

The test is

$$M_y = N_x$$

on a suitable simply connected region. Then the solution is

$$\Phi(x,y) = C.$$

If the equation is not exact, one may look for an integrating factor, often depending only on x or only on y.

§1.3 Higher-Order ODEs and Characteristic Roots

If a second-order equation does not explicitly contain y, set $p = y'$ and reduce the order. If it does not explicitly contain x, set $p = p(y) = y'$, so

$$y'' = \frac{dp}{dx} = \frac{dp}{dy}\frac{dy}{dx} = p\frac{dp}{dy}.$$

These substitutions are not random: they use a missing variable as a symmetry.

A second-order linear equation has

$$y'' + p(x)y' + q(x)y = f(x).$$

The associated homogeneous equation has a solution space. If y_1, y_2 are independent homogeneous solutions, then

$$y_h = C_1 y_1 + C_2 y_2.$$

A general nonhomogeneous solution is

$$y = y_h + y_p.$$

This is the affine-space principle: one particular solution plus the kernel of a linear operator.

The Wronskian

$$W(y_1, y_2) = y_1 y_2' - y_1' y_2$$

measures independence. If it is nonzero at one point for solutions of a linear homogeneous equation, the two solutions form a fundamental system.

For

$$y'' + ay' + by = 0,$$

try $y = e^{rx}$. The characteristic equation is

$$r^2 + ar + b = 0.$$

Distinct real roots r_1, r_2 give

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}.$$

A repeated root r gives

$$y = (C_1 + C_2 x)e^{rx}.$$

Complex roots $\alpha \pm i\beta$ give

$$y = e^{\alpha x}(C_1 \cos \beta x + C_2 \sin \beta x).$$

For

$$L[y] = f(x), \qquad L = D^2 + aD + b,$$

the method of undetermined coefficients works when f is built from polynomials, exponentials, sines, and cosines. If the trial function overlaps with a homogeneous solution, multiply by a sufficient power of x.

Variation of parameters works more generally. If y_1, y_2 are fundamental solutions, seek

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x).$$

The auxiliary condition $u_1' y_1 + u_2' y_2 = 0$ produces a solvable linear system for u_1', u_2'.

§1.4 Initial and boundary conditions select solutions

An initial-value problem specifies

$$y(x_0) = a, \qquad y'(x_0) = b.$$

For regular linear equations, existence and uniqueness say that these data select exactly one solution. This is conceptually important: a second-order equation needs two pieces of initial data because its solution space is two-dimensional.

§1.5 Differential operators and response kernels

Let

$$L[y] = y'' + p(x)y' + q(x)y.$$

Then a differential equation is an operator equation

$$L[y] = f.$$

The homogeneous solutions form $\ker L$. A nonhomogeneous solution set, if nonempty, is an affine translate of $\ker L$. Initial conditions are linear functionals that choose one element from that affine space. Thus a linear differential equation resembles a linear system, but the domain of the differential operator and its boundary conditions must be specified before asking whether an inverse exists. The Wronskian plays the local role of a determinant for fundamental solutions.

For a linear operator L, a Green function $G(x, \xi)$ is a kernel that represents the inverse problem. Formally,

$$L_x G(x, \xi) = \delta(x - \xi),$$

where δ is the Dirac delta distribution. Then a solution of

$$L[y] = f$$

can often be written as

$$y(x) = \int G(x, \xi) f(\xi)\, d\xi$$

with boundary conditions built into G.

The boundary conditions determine the kernel. For the simple Dirichlet problem $-y'' = f$ on $(0, 1)$, with $y(0) = y(1) = 0$, one can calculate

$$G(x, \xi) = \begin{cases} x(1 - \xi), & x \le \xi, \\ \xi(1 - x), & x \ge \xi. \end{cases}$$

It vanishes at the endpoints; its first derivative jumps by -1 at $x = \xi$, so $-\partial_x^2 G = \delta_\xi$ distributionally. Hence $y(x) = \int_0^1 G(x, \xi) f(\xi)\, d\xi$ solves the boundary-value problem for sufficiently regular forcing. This is the precise operator-theoretic form of superposition.

For first-order evolution equations, one often writes

$$\frac{du}{dt} = Au.$$

If A were a number, the solution would be $u(t) = e^{tA}u(0)$. In a Banach or Hilbert space, A may be an unbounded operator. If A is a generator of a strongly continuous semigroup, the solution (in the appropriate mild or strong sense) is described by a semigroup

$$T(t) = e^{tA}, \qquad T(t + s) = T(t)T(s).$$

The heat equation is the model example: the Laplacian generates a smoothing semigroup. The elementary exponential solution of constant-coefficient ODEs is the finite-dimensional shadow of this theory.

A second-order operator of the form

$$L[y] = -(p(x)y')' + q(x)y$$

with suitable boundary conditions is often self-adjoint. Its eigenvalue problem

$$L[y] = \lambda w(x) y$$

produces orthogonal eigenfunctions with respect to the weighted inner product

$$\langle u, v \rangle_w = \int_a^b u(x) v(x) w(x)\, dx,$$

provided $p > 0$, $w > 0$, and self-adjoint separated boundary conditions hold. Integration by parts then gives

$$(\lambda_m - \lambda_n) \int_a^b w u_m u_n\, dx = \big[p(u_m u_n' - u_m' u_n)\big]_a^b.$$

The boundary term vanishes under self-adjoint boundary conditions, forcing orthogonality for distinct eigenvalues. Fourier sine and cosine series are special cases of this principle.

Thus the elementary method of solving $y'' + \lambda y = 0$ under boundary conditions is not an isolated trick. It is the beginning of spectral theory for differential operators.